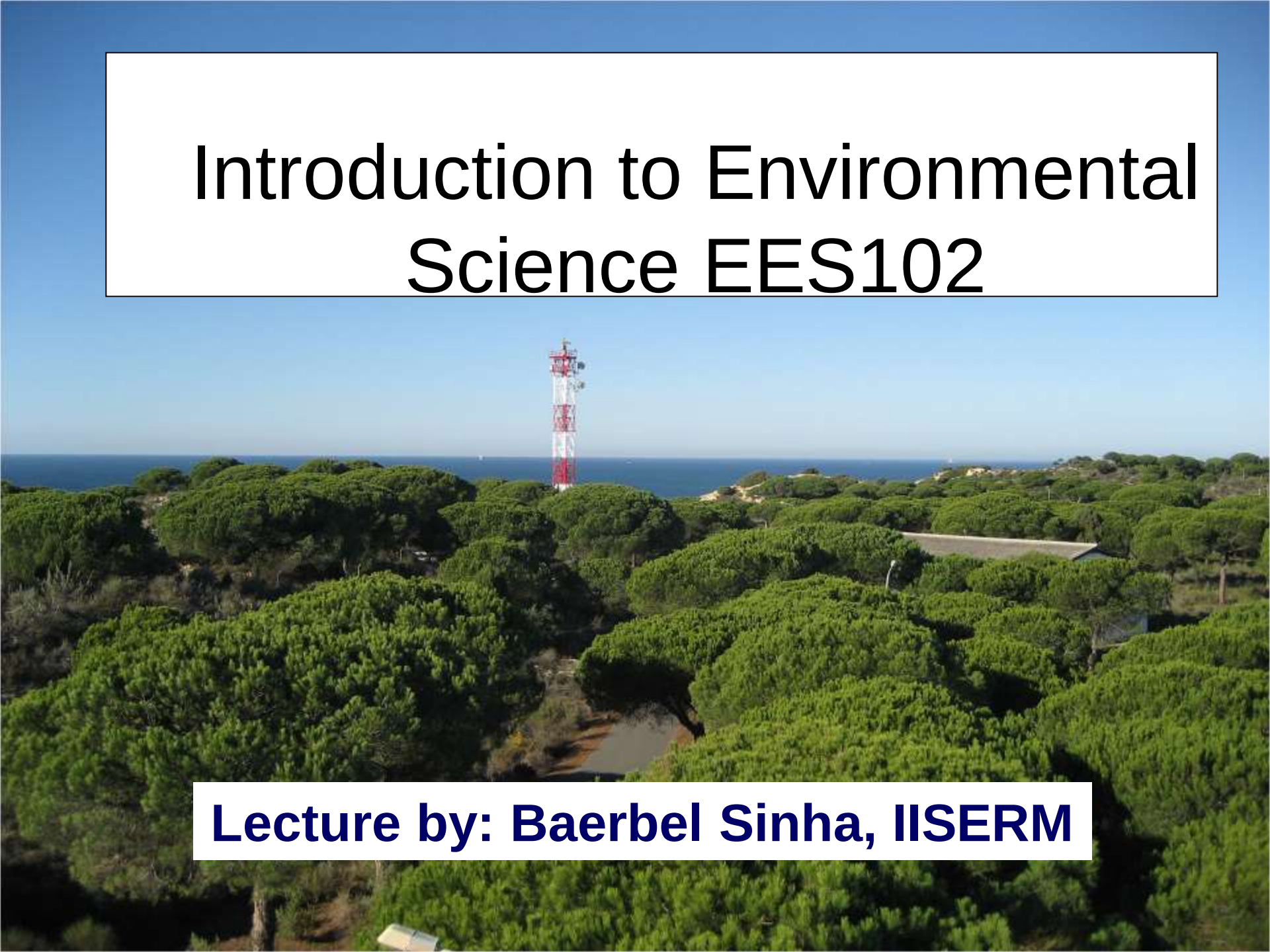


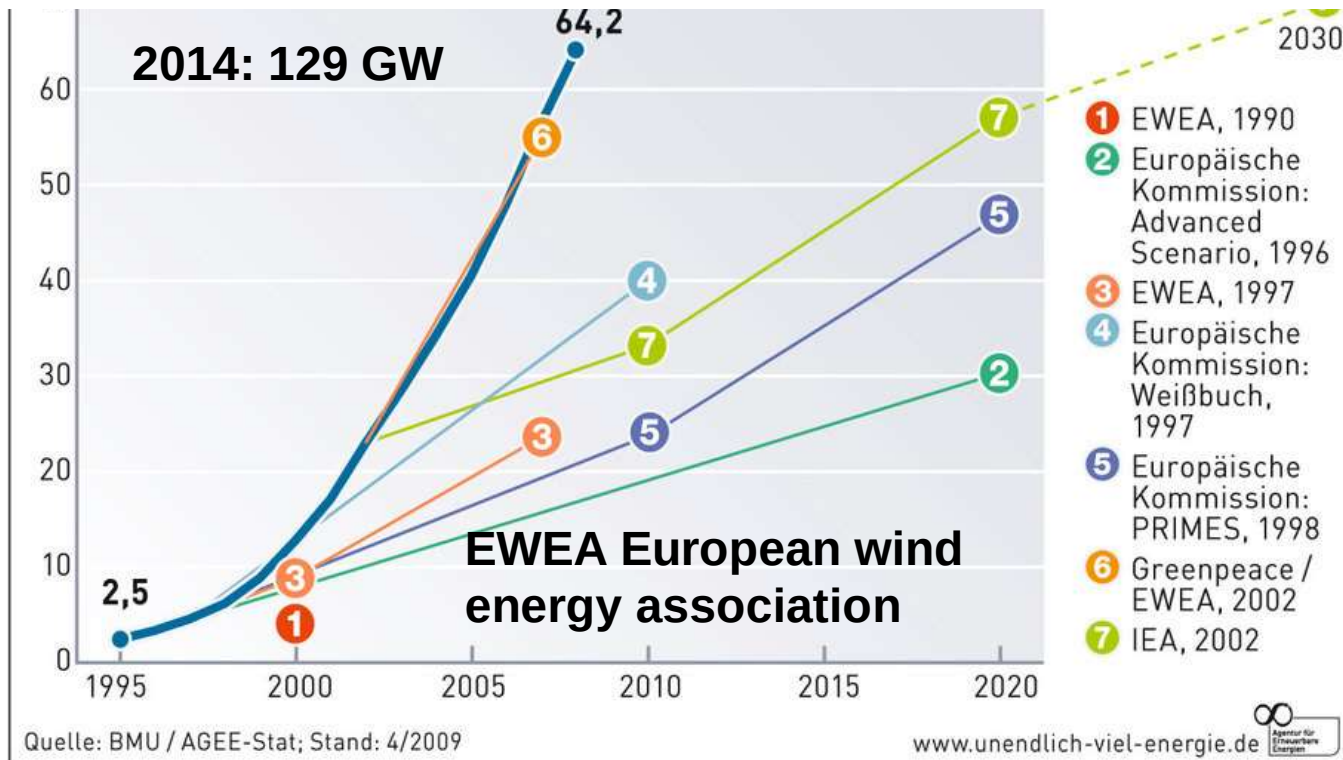
Introduction to Environmental Science EES102



Lecture by: Baerbel Sinha, IISERM

Prognosis and reality

While wind power was becoming big and cheap, several agencies published estimates about how much it would contribute to the European electricity supply till 2020. The bold blue curve shows the **real development till 2008**. Note that the European Commission and the international Energy Agency never came close to capturing the real trend even for projections published as late as 2002. **Orders for wind parks that started feeding the grid in 2005-2006 HAD ALREADY BEEN PLACED BY THEN! Only the Greenpeace funded study captured the actual trend but their projection was discredited as biased**, although it was simply based on published order books from stock-market reporting major wind mill manufacturers!



If you want real news about what is going on with respect to renewables and the energy transition read “Business Insider”, “Forbes”, and quarterly stock market reports of major companies

CLIMATE CHANGE MITIGATION

Climate change mitigation consists of actions to limit the magnitude or rate of global warming and its related effects. This generally involves reductions in human emissions of greenhouse gases (GHGs) and short lived climate forcers (SLCFs)

Fossil fuels account for about 70% of GHG emissions. The main challenge is to eliminate the use of coal, oil and gas and substitute these fossil fuels with clean energy sources.

Due to massive price drops, wind power and solar photovoltaics (PV) are increasingly out-competing oil, gas and coal though these require energy storage and extended electrical grids.

Mitigation or reversal of climate change, may also be achieved by replacing petrol and diesel with electric vehicles, as vehicles are replaced roughly every 10-20 years and hence complete replacement of certain vehicle classes before 2040 appears to be economically and technically feasible.

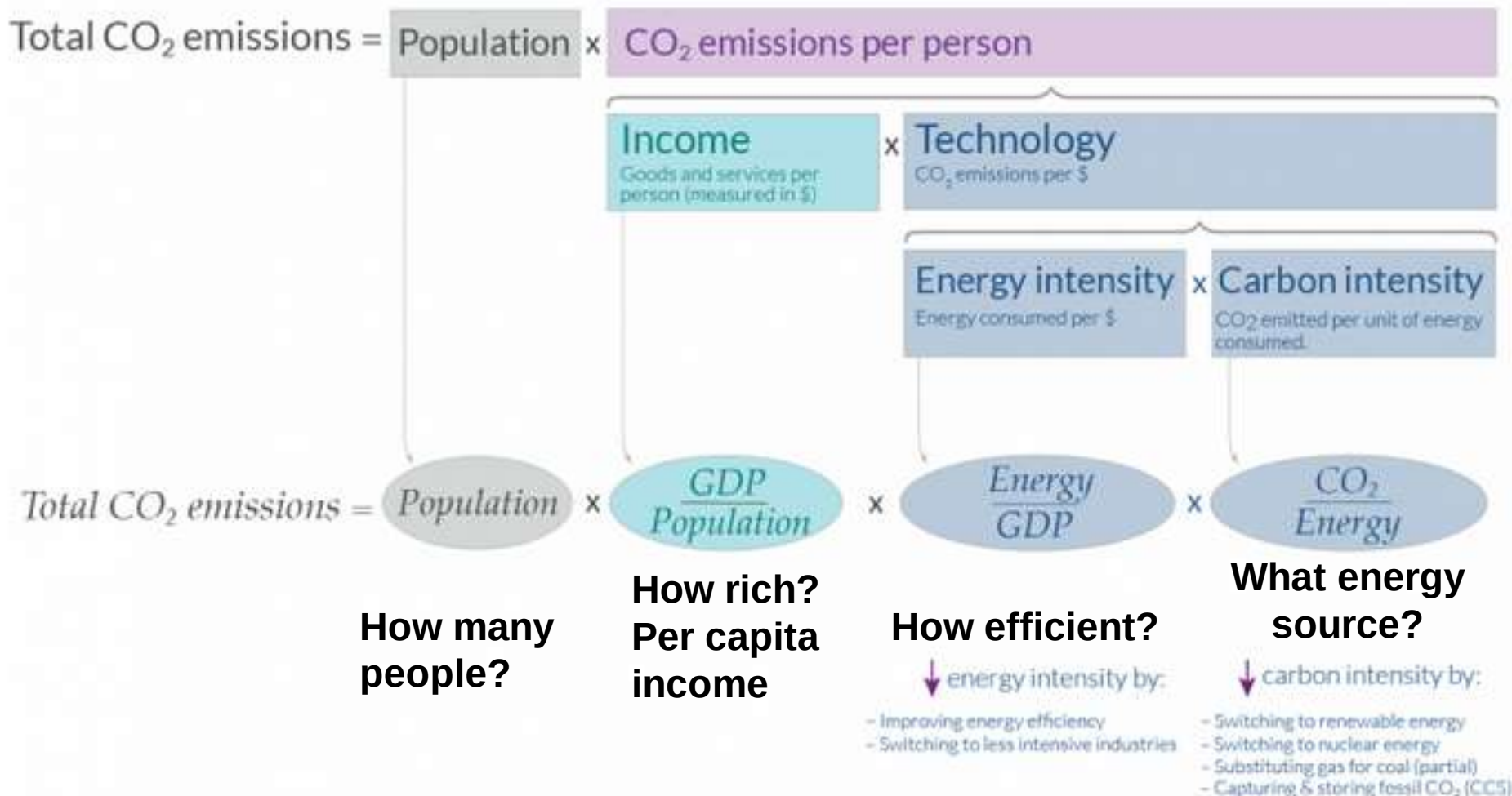
Reforestation and forest preservation (a "carbon sink"), changes to agriculture practice and machinery, divestment from fossil fuel finance, democratic corporate governance reforms, changes to consumer laws, and implementing a green recovery after the COVID-19 pandemic can also help mitigate climate change.

KAYA IDENTITY OR WHAT NEEDS TO CHANGE?

What determines total CO₂ emissions?

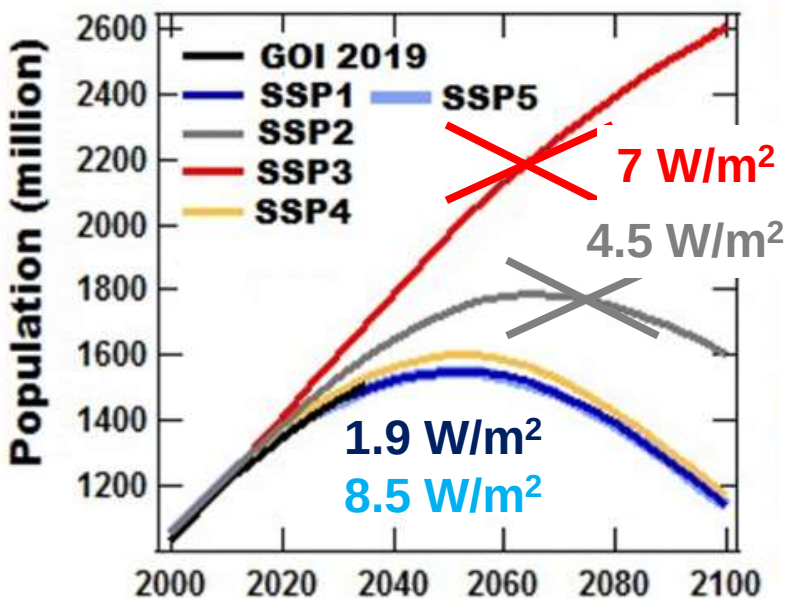
Our World
in Data

The 'Kaya Identity' breaks down total emissions into the key elements driving them.

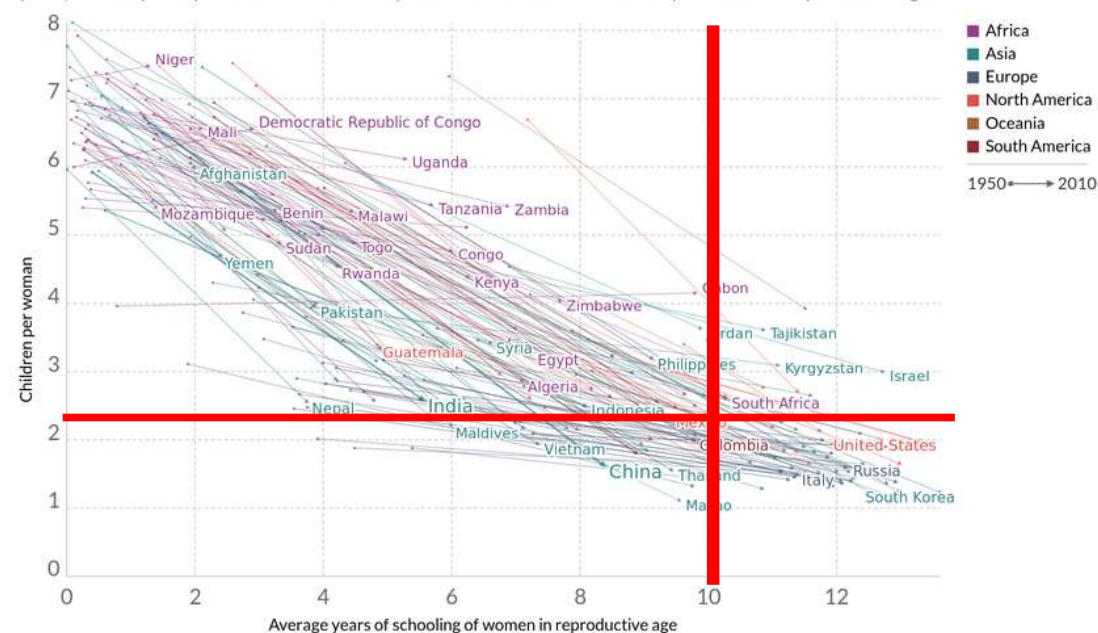


EDUCATING GIRLS AND FAMILY PLANNING: INDIA'S BIGGEST ACCOMPLISHMENT THE RTE

- ▶ Number #6. & #7. in the list of the most efficient ways to combat climate change <https://drawdown.org/>
- ▶ The RTE has done it's job. Population will peak @ less than 1.55 billion



Women's educational attainment vs. number of children per woman
Shown on the x-axis is the average number of years of schooling of women in the reproductive age (15 to 49 years). On the y-axis you find the 'total fertility rate' – the number of live births per woman in reproductive age.



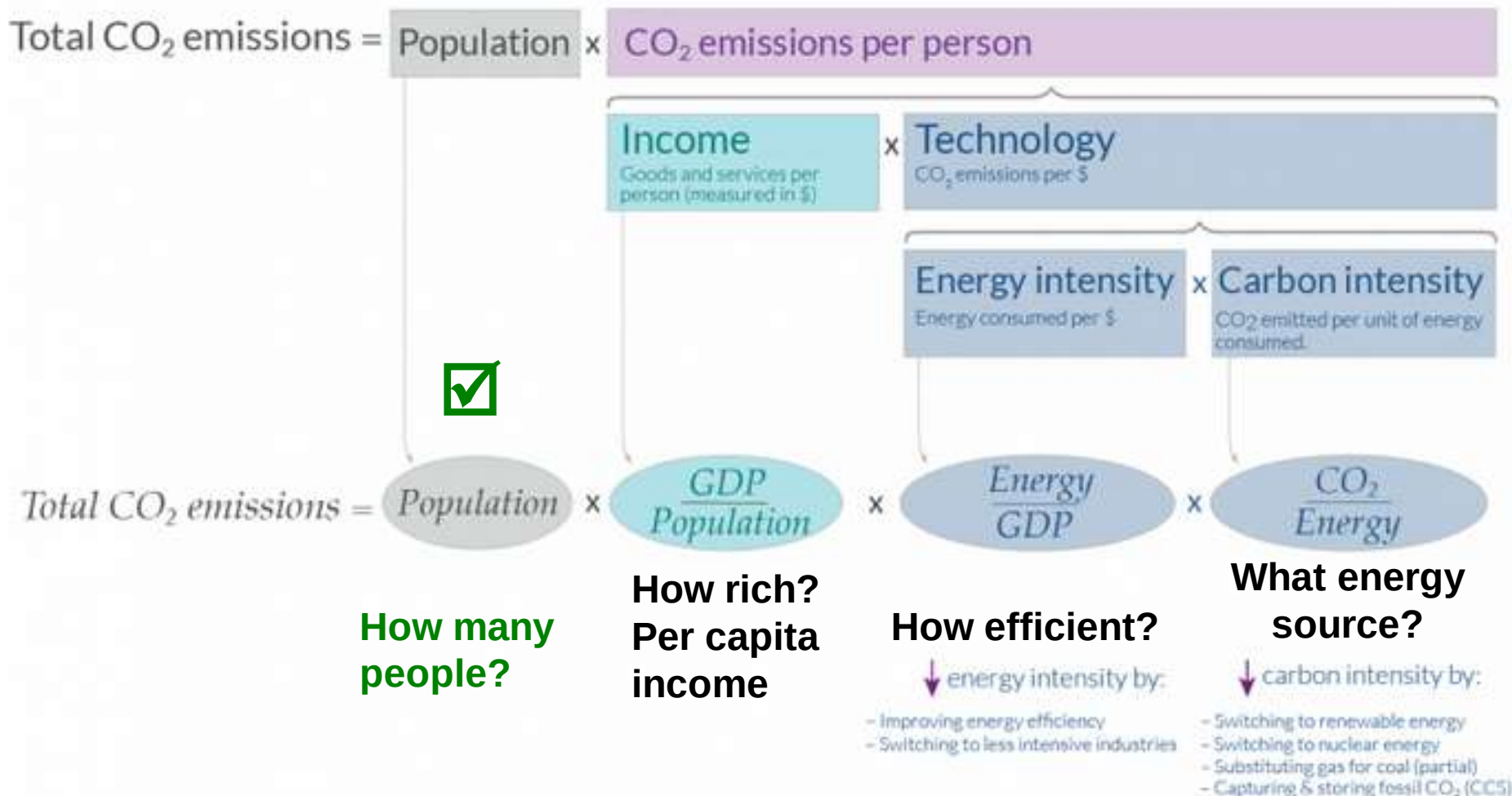
Source: United Nations – Population Division (2019 Revision), Our World In Data (2017)

KAYA IDENTITY OR WHAT NEEDS TO CHANGE?

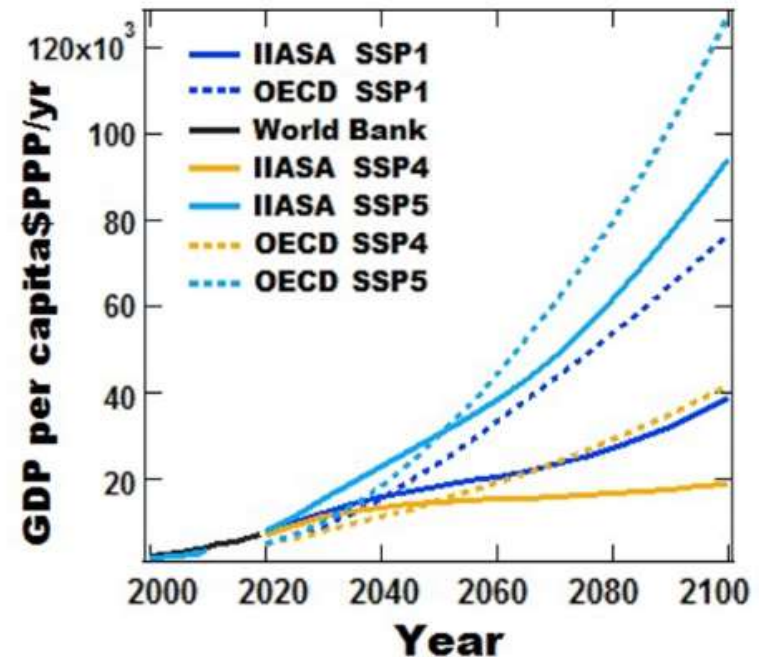
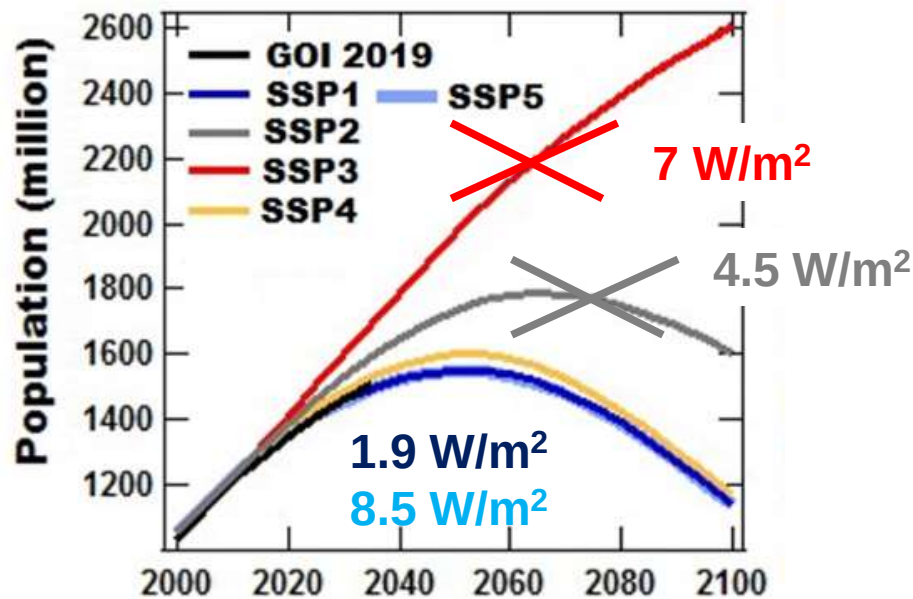
What determines total CO₂ emissions?

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SO WHERE IS INDIA RIGHT NOW?

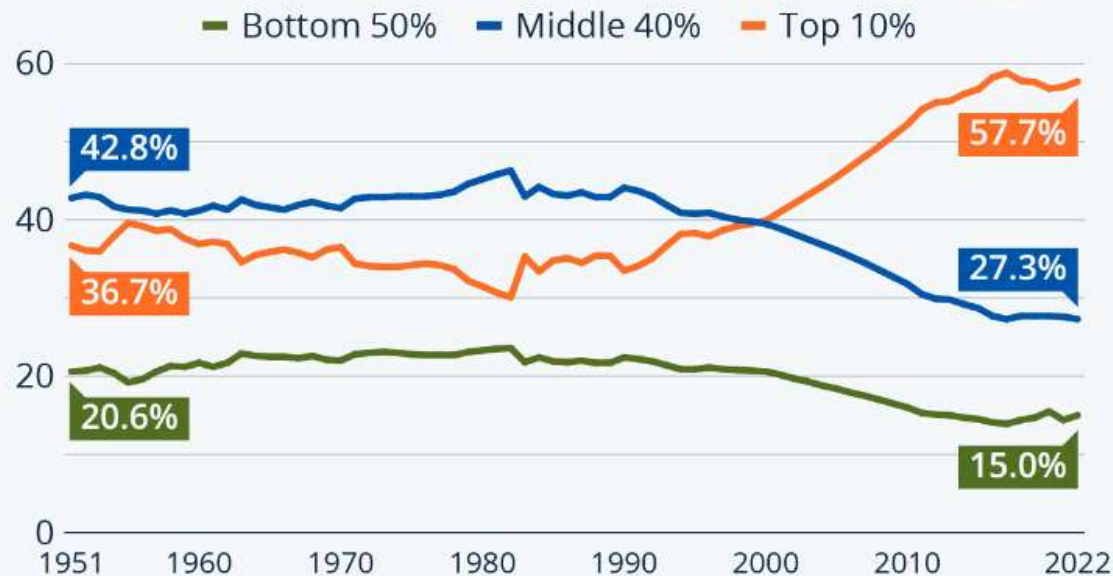


- The economists are adamant that fossil fuelled economic growth is faster and leads to higher incomes & living standards ... however, that is yet to be proven
- Several countries with a >90% renewable energy system are very prosperous e.g. Norway
- Anyone that proclaims that India needs to reach the same per capita energy usage developed nations had in 2010 is insisting that we should be sticking with incandescent bulbs and other old inefficient technology till 2040 instead of buying the best the market currently has on offer

INCOME INEQUALITY IS THE DIFFERENCE BETWEEN SSP 4 AND THE OTHER TWO

How the Income of India's Richest 10% Surged

Per-adult pre-tax national income shares of different income groups in India (in percent)



Authors' estimates combine national income account aggregates, tax tabulations, and income and consumption surveys.

Source: World Inequality Lab

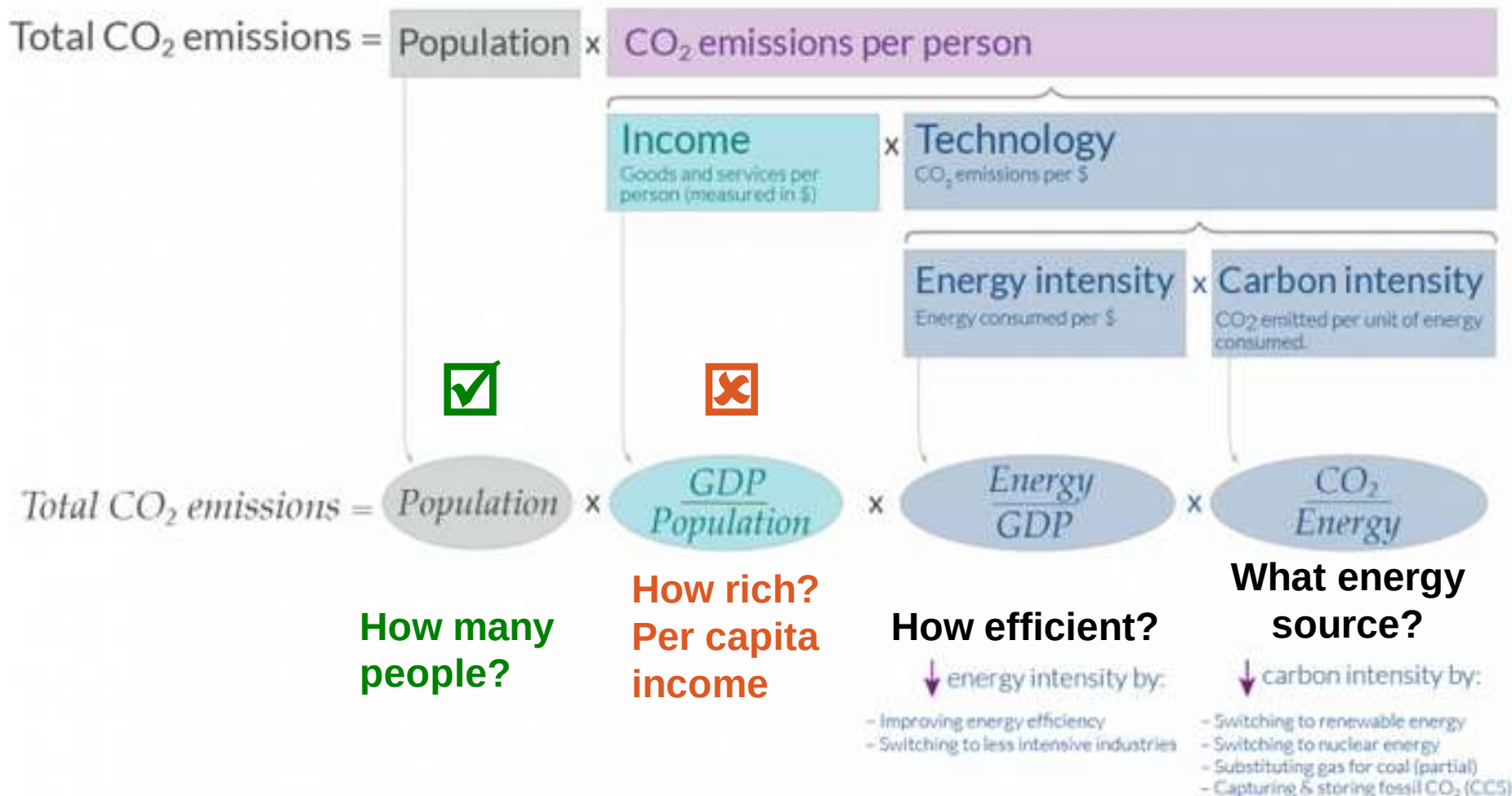


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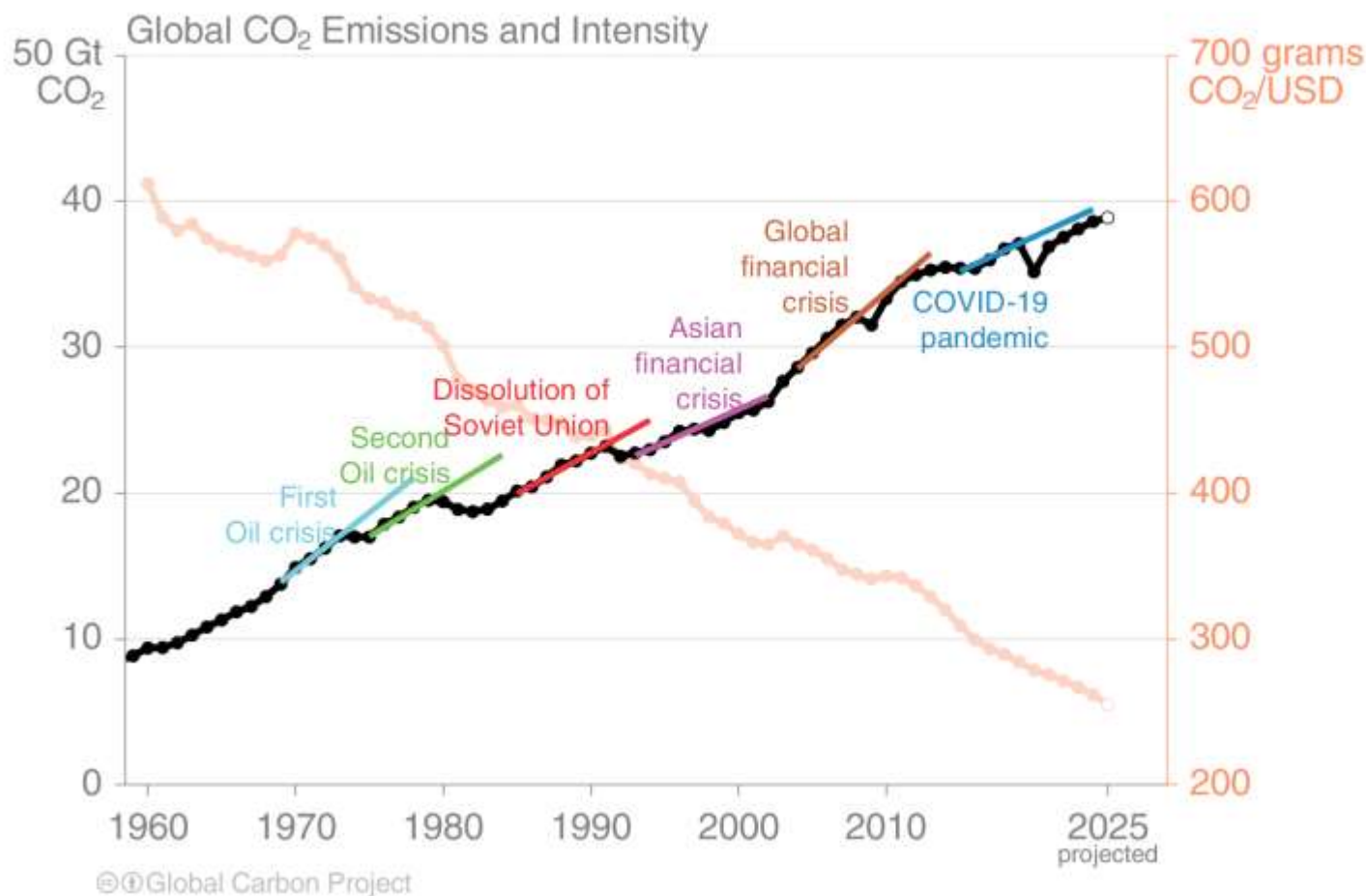
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FOSSIL CO₂ EMISSION INTENSITY

Global CO₂ emissions growth has generally resumed quickly from global crises. Emission intensity has steadily declined but not sufficiently to offset economic growth.



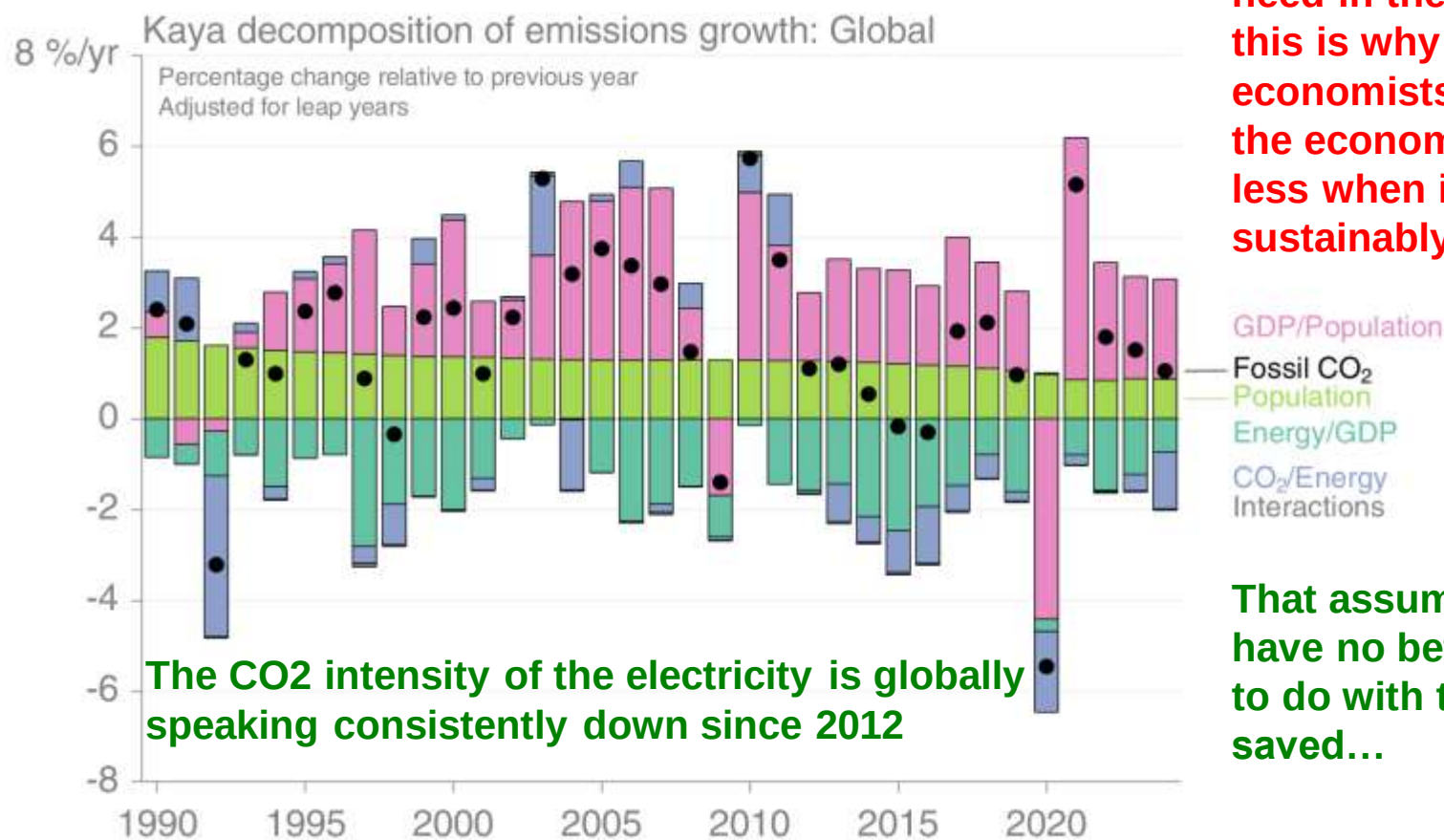
Each trend line is based on the five years before the crisis and extended to five years after. Economic activity is measured in purchasing power parity (PPP) terms in 2017 US dollars.

Source: [Friedlingstein et al 2025](#); [Global Carbon Project 2025](#)

FOSSIL CO₂ EMISSIONS — KAYA DECOMPOSITION

Globally, decarbonisation and declines in energy per GDP are largely responsible for the reduced growth rate in emissions over the last decade. 2020 was a clear outlier with a sharp decline in GDP.

The cheapest energy is the one you don't need in the first place this is why economists assume the economy grows less when it grows sustainably

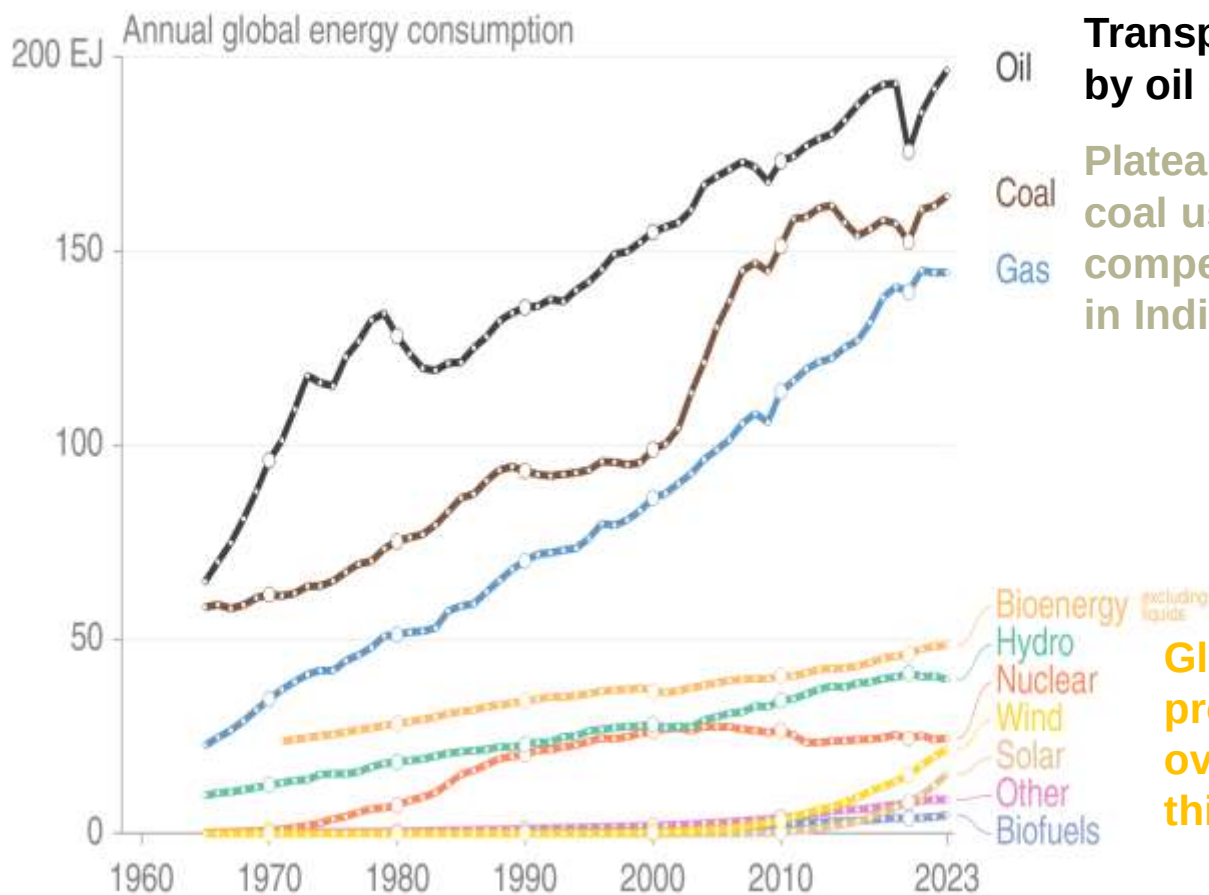


The CO₂ intensity of the electricity is globally speaking consistently down since 2012

That assumes you have no better things to do with the money saved...

ENERGY USE BY SOURCE

Consumption of natural gas was flat in 2023, but oil exceeded its pre-pandemic level.
Renewable energy continued to grow, but needs to grow even faster to replace fossil energy consumption.



Transport is still driven by oil - globally

Plateau reductions of coal use in US & EU compete with growth in India & China

Global wind energy production will likely overtake nuclear this year

© Global Carbon Project • Data: Energy Institute, IEA (bioenergy)

This figure shows “primary energy” using the substitution method
(non-fossil sources are scaled up by an assumed fossil efficiency of approximately 0.38)

Source: [Energy Institute 2024](#); [Global Carbon Project 2024](#)

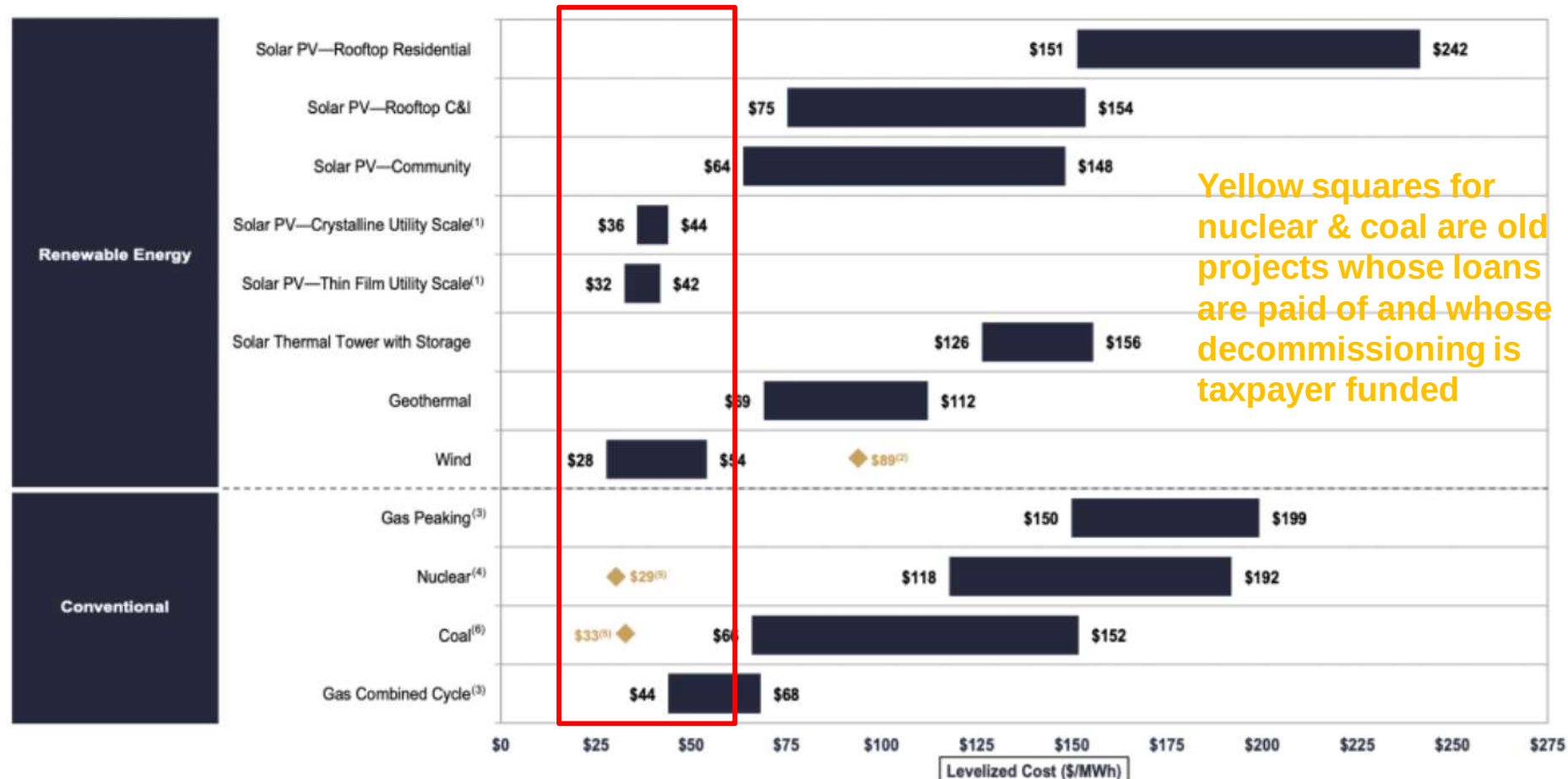
NUCLEAR & COAL ARE NOT COST COMPETITIVE – ONLY TAXPAYER GUARANTEED LOANS POSSIBLE NOWADAYS

LAZARD

LAZARD'S LEVELIZED COST OF ENERGY ANALYSIS—VERSION 13.0

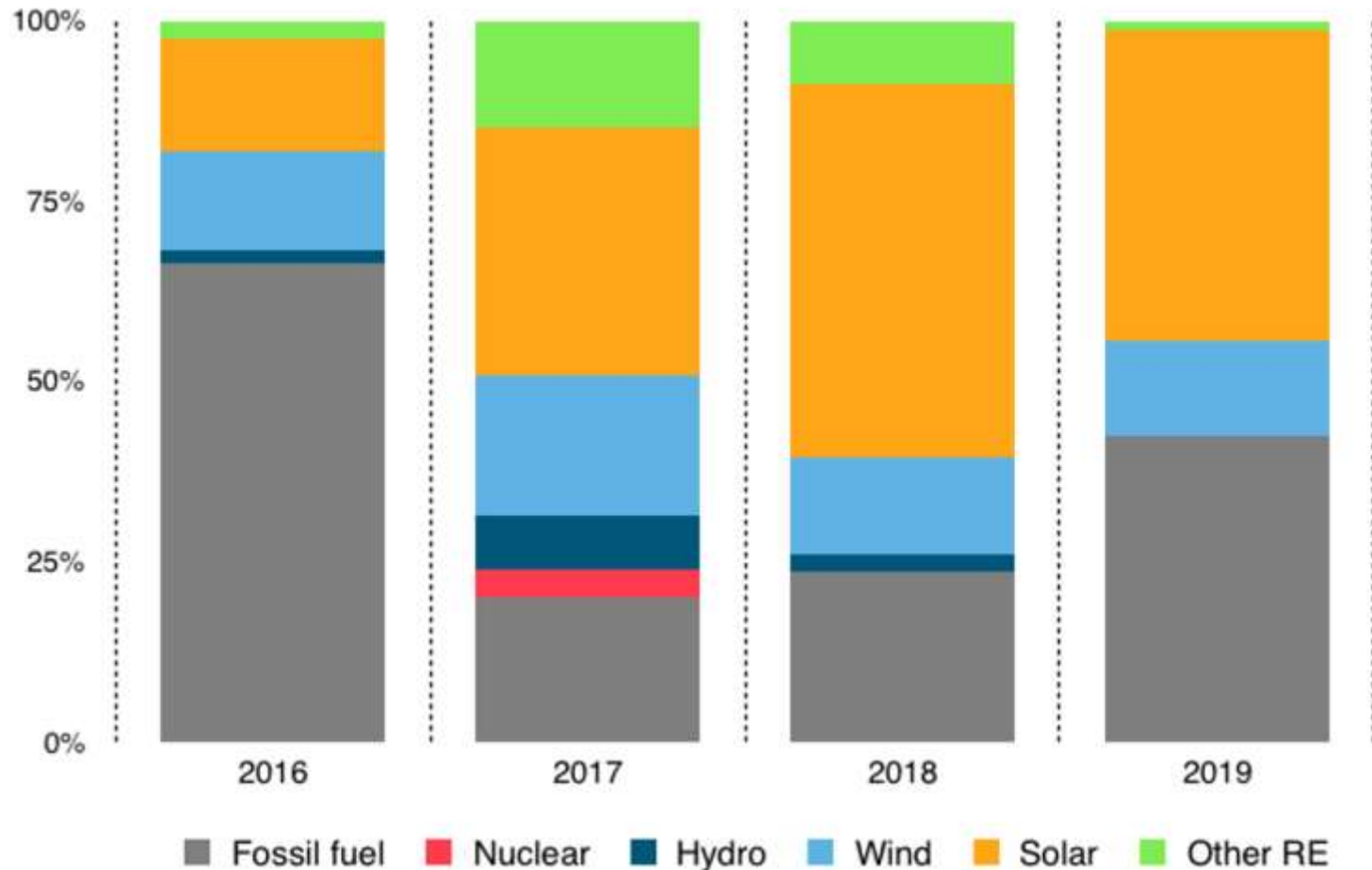
Levelized Cost of Energy Comparison—Unsubsidized Analysis

Selected renewable energy generation technologies are cost-competitive with conventional generation technologies under certain circumstances



ENERGY SOURCES WIND AND SOLAR: ECONOMICS IS WINNING – RENEWABLES ARE BEATING FOSSIL FUELS IN FRESHLY COMMISSIONED CAPACITY

Share of technologies in annual capacity addition in India



Data: Government of India

India has 35% renewables at present

SOLAR MAKES THE EVENING NECK WORSE... THAT IS WHY WE LAUNCHED UJALA

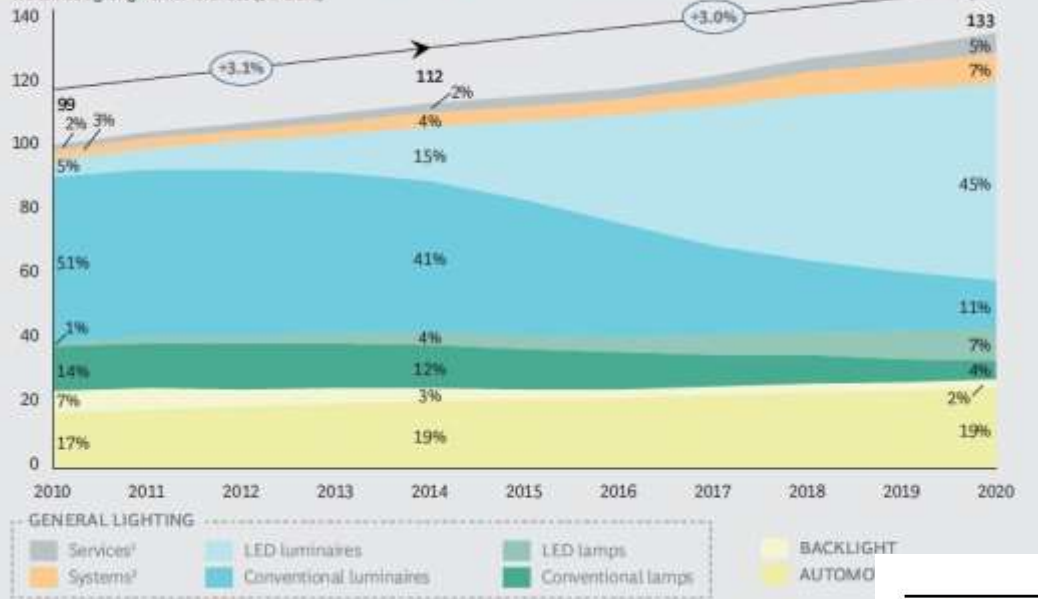
Expected All India Duck Curve
(After considering 20000 MW of Solar Generation by 2021-22)



LED bulbs cut load shedding because they reduce the height of the evening neck which solar, coal & nuclear can't supply

The Changing Composition of the Lighting Market

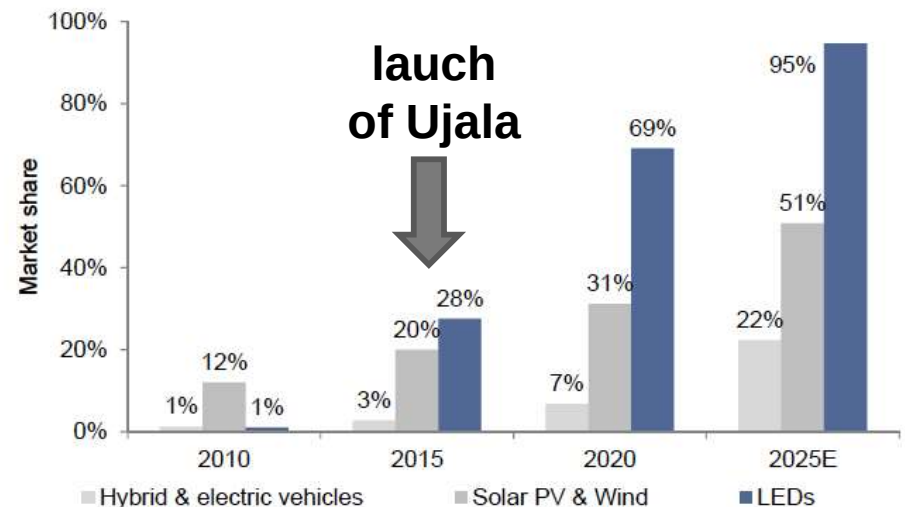
Worldwide lighting-market revenues (\$ billions)



Ujala speeded up global LED adoption

India became the world's LED capital, accounting for 10-12% of the global 2018 market of 4 billion bulbs

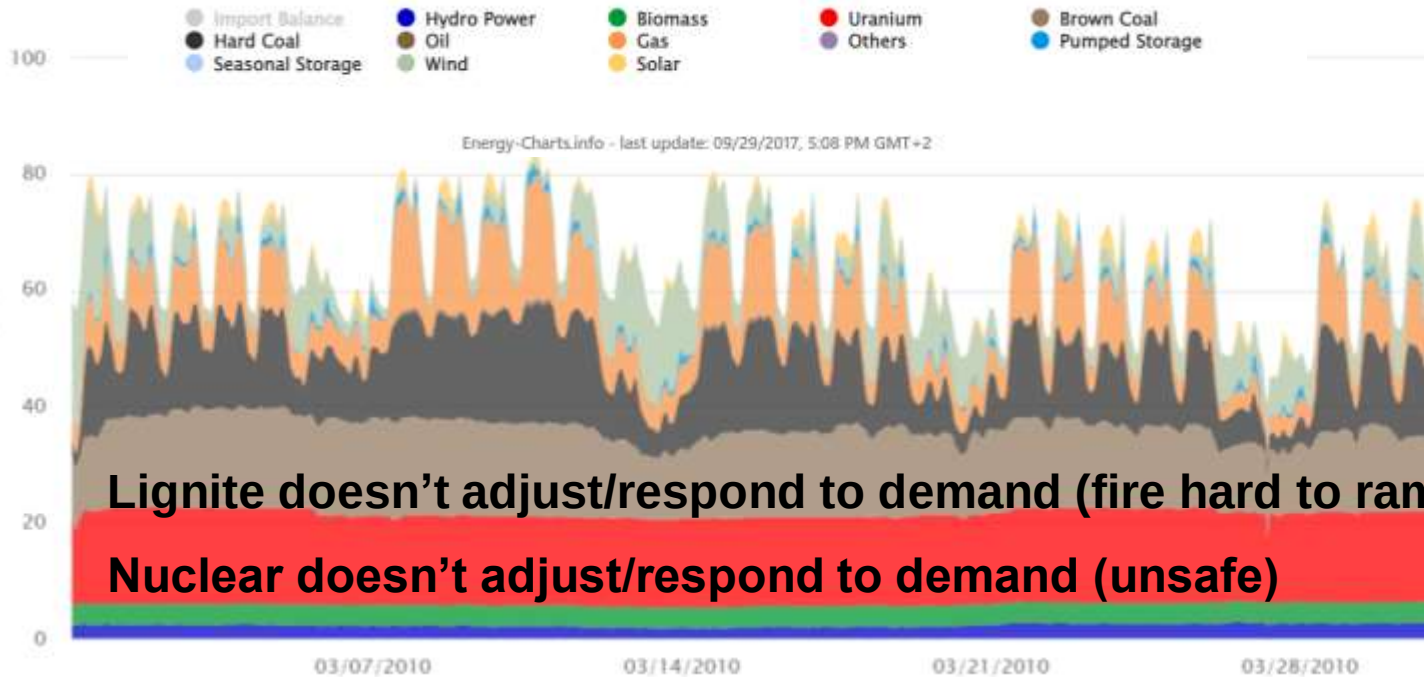
Exhibit 19: The rapid adoption of LEDs in lighting marks one of the fastest technology shifts in human history. LEDs for general lighting were commercialized only recently, but will dominate sales by the end of this decade



Source: IHS, Company data, IEA, IRENA, BP, Goldman Sachs Global Investment Research.

POWER SECTOR 1.1.: WHY A GRID BASED ON LIGNITE COAL + NUCLEAR WORKS ONLY WITH LOAD SHEDDING

Electricity production in Germany in March 2010



Gas does the adjustments

Hard coal can be adjusted but is costly & imported

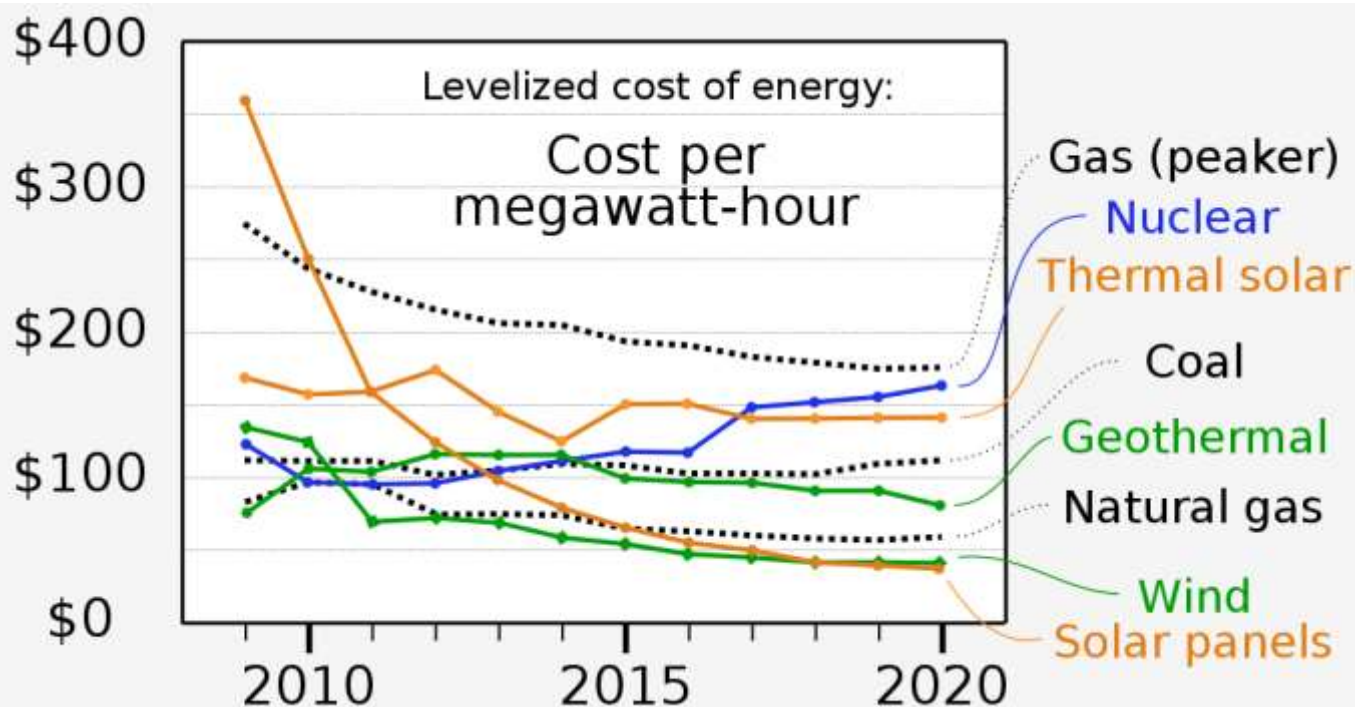
Lignite doesn't adjust/respond to demand (fire hard to ramp up down)

Nuclear doesn't adjust/respond to demand (unsafe)

Without gas stations that supply peak demand only (of which India has few), or expensive high tech hard coal power plants load shedding is inevitable...

WHY NUCLEAR CAN'T RESPOND TO DEMAND FLUCTUATIONS AND IS A BAD IDEA:

1. If you want to know what happens if nuclear is ramped down without following a complicated security protocol then google Chernobyl disaster. Making nuclear respond to demand fluctuations means testing your luck too often.
2. It's the costliest form of non-responsive base power on the market!

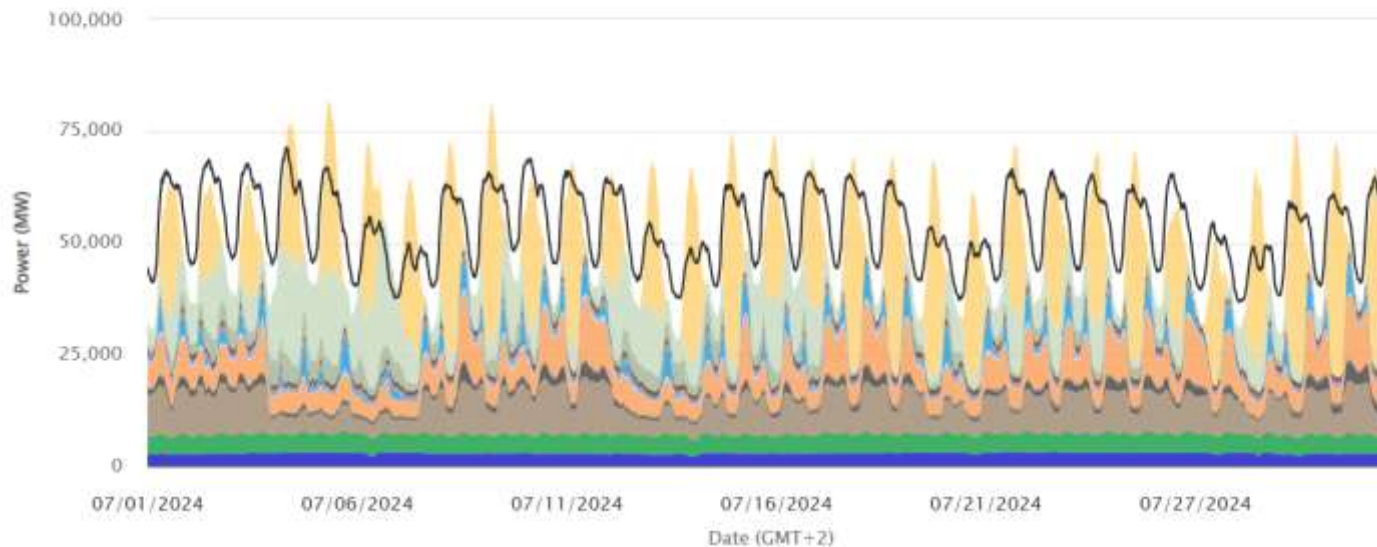


SOLAR REPLACES (IMPORTED) GAS, (IMPORTED) HARD COAL AND DAYTIME LOAD SHEDDING

Legend: Import Balance, Hard Coal, Seasonal Storage, Hydro Power, Oil, Wind, Biomass, Gas, Solar, Uranium, Others, Brown Coal, Pumped Storage

Total net electricity generation in Germany in July 2024

Energetically corrected values



**Solar reduced
need for daytime
adjustment**

**Pumped storage
can take some gas
load**

**Lignite has been
retrofitted to be
more adjustable**

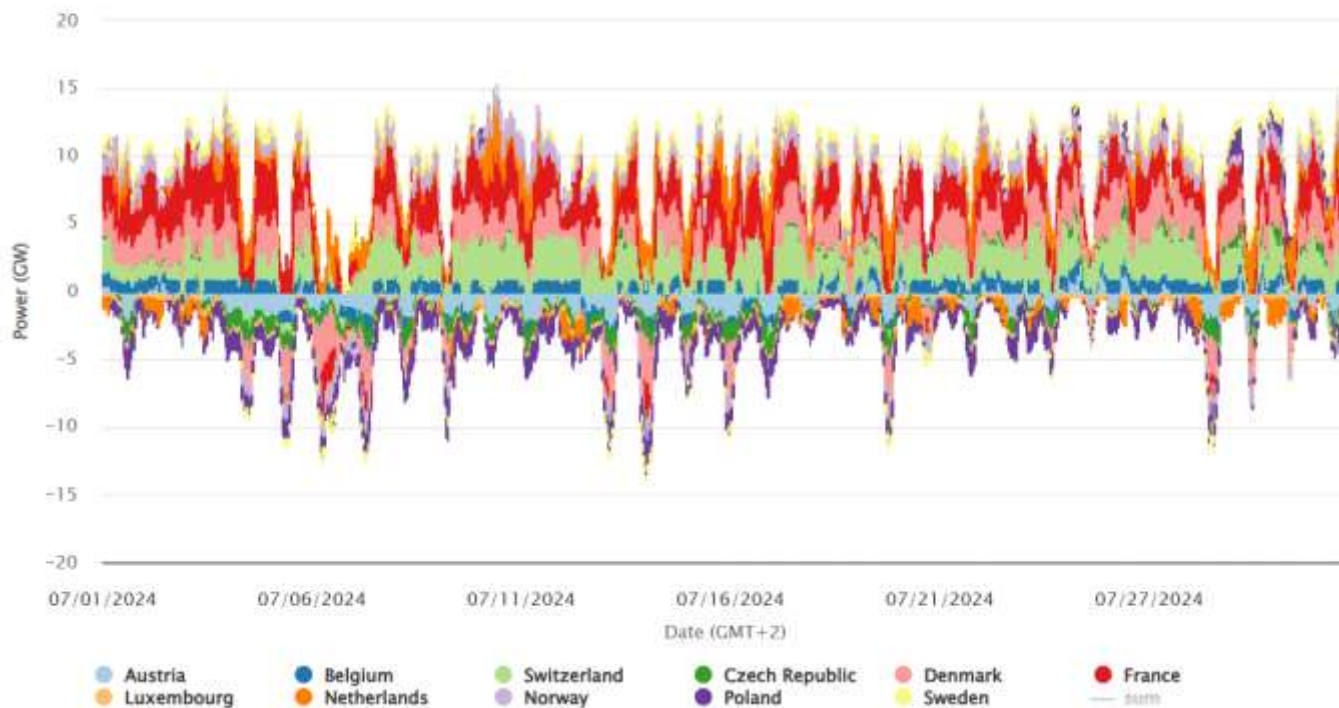
**>50% renewable
grid without
sufficient storage
capacity becomes
import/export
dependent**

Solar reduces the need for gas. In the Indian context solar reduces the need for daytime load shedding of peak AC loads. More solar can also take care of electrical mid day meal cooking (reason for the high mid day load in Germany almost 100% electric cooking)

SOLAR REPLACES (IMPORTED) GAS, (IMPORTED) HARD COAL AND DAYTIME LOAD SHEDDING

Cross border electricity trading of Germany in July 2024

Positive values indicate import. Negative values indicate export.



>50% renewable grid without sufficient storage capacity becomes import/export dependent

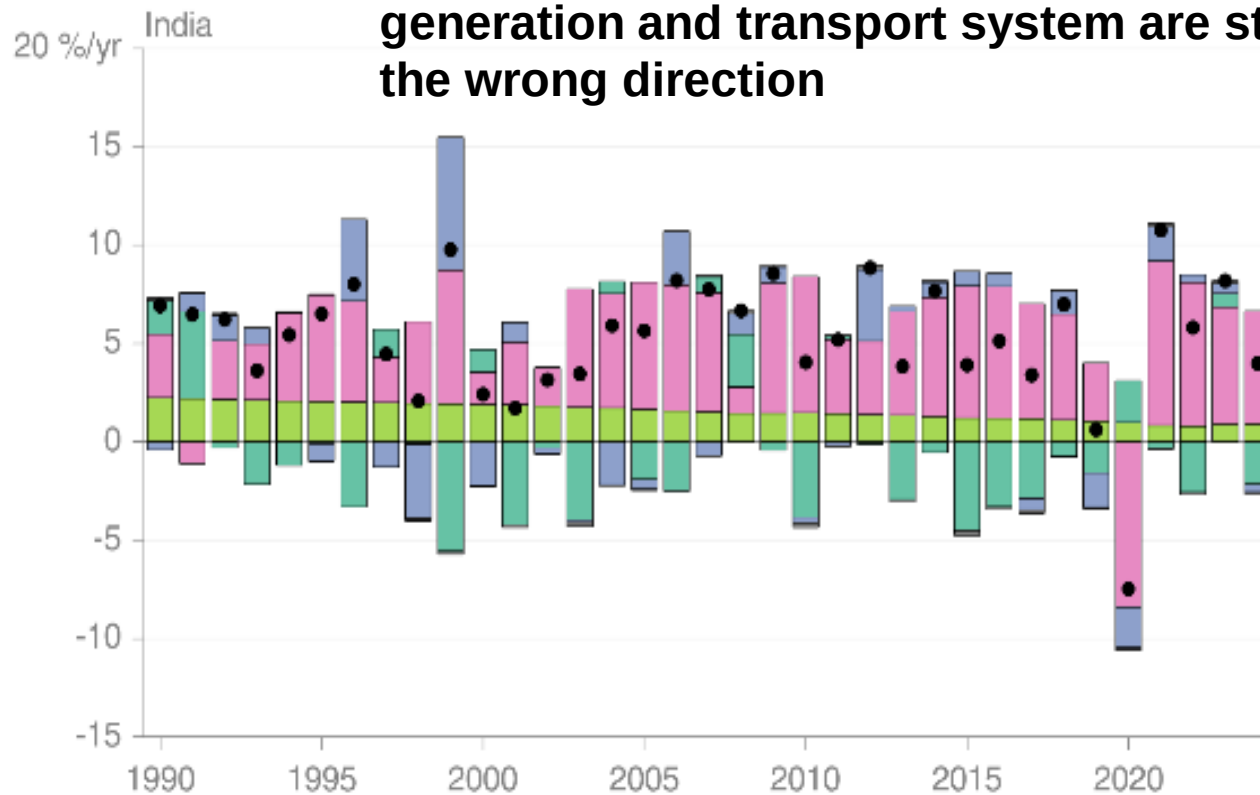
Excess solar is exported to Sweden/Norway (cheap pumped hydro storage) and reimported at night

France imports limited amounts of excess solar but exports nuclear at night (remember nuclear only good for base load). France has about 70% nuclear generation. Denmark has more offshore wind and exports at night. Switzerland has lot of hydro including pumped hydro.

FOSSIL CO₂ EMISSIONS — KAYA DECOMPOSITION

Globally, decarbonisation and declines in energy per GDP are largely responsible for the reduced growth rate in emissions over the last decade. 2020 was a clear outlier with a sharp decline in GDP.

In India the carbon intensity of the electricity generation and transport system are still moving in the wrong direction

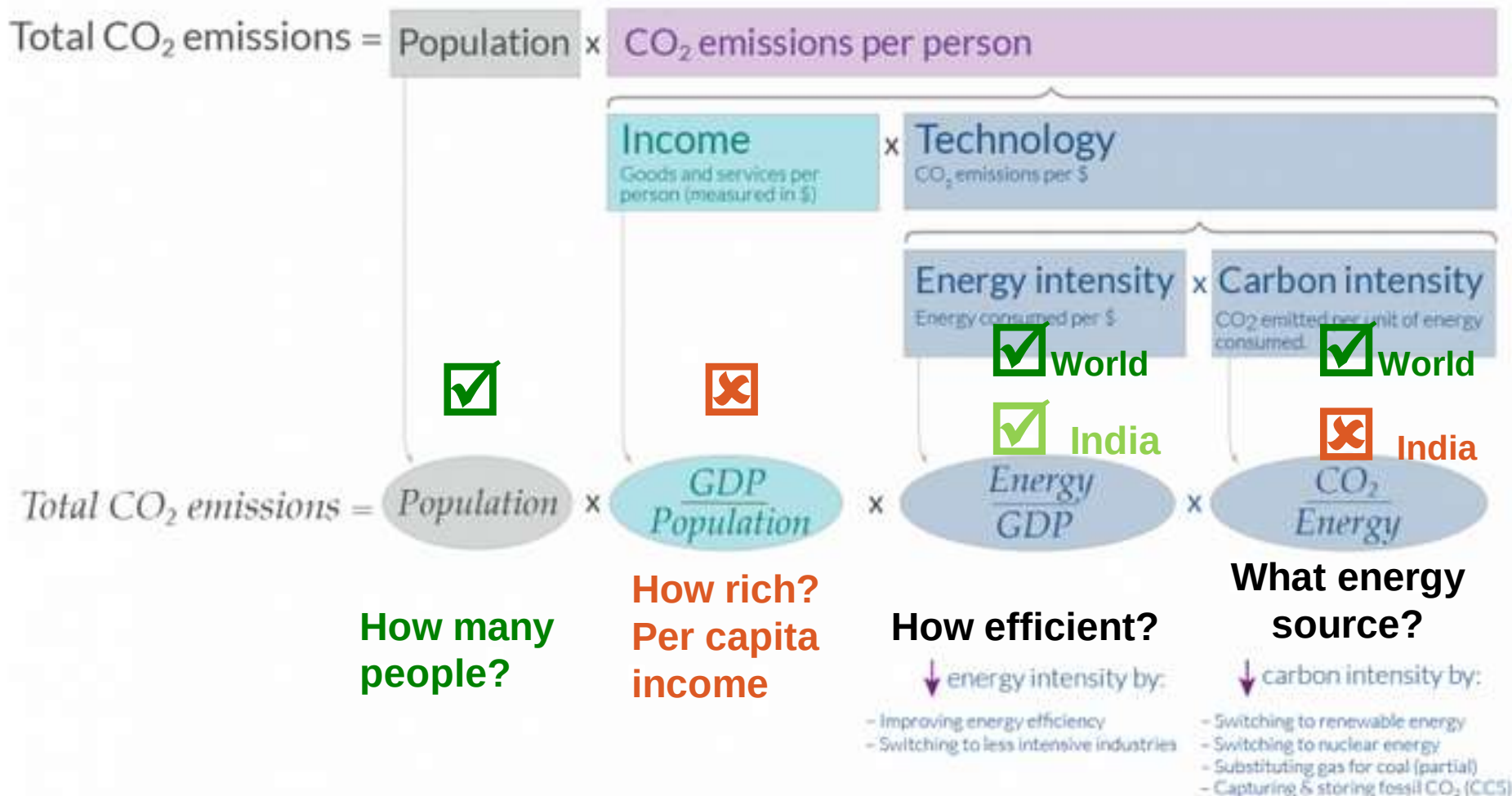


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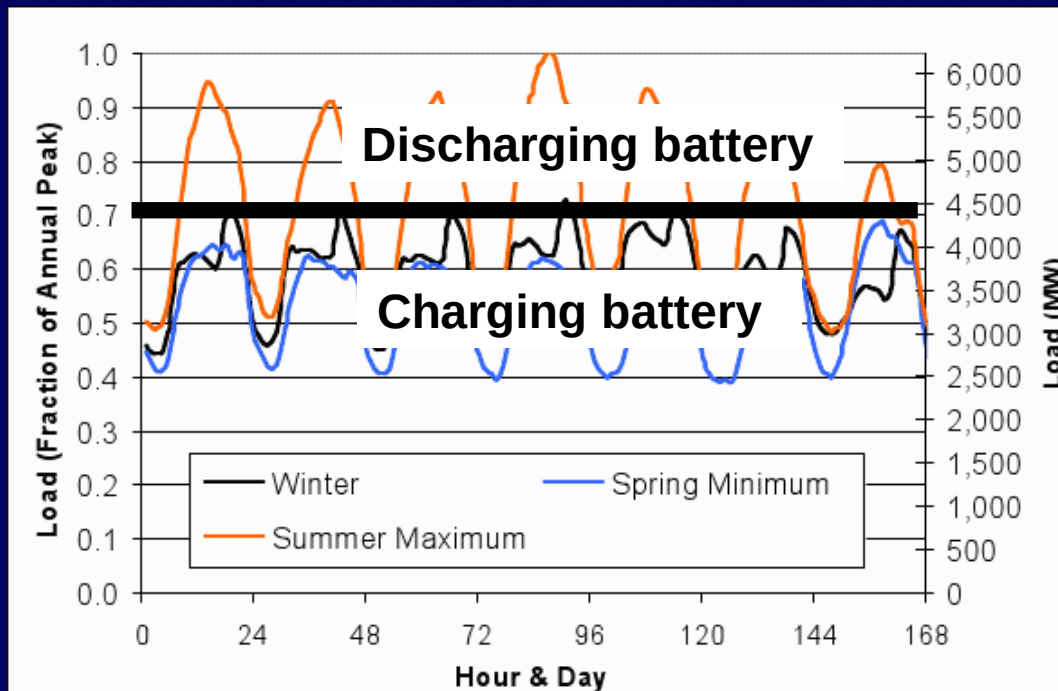
DISRUPTIVE TECHNOLOGY: POWER CONSUMPTION

The last disruptive technology from a power consumption viewpoint were LED's they cut the CO₂ footprint of lighting purposes by a factor of 20-25. Ujala scheme forced their rapid adoption in India and forced the prices down through massive tenders. Many developing countries are now buying Indian LED bulbs. Just the LED bulbs distributed under Ujala cut the evening peak demand by 9.5 GW

Air heat pumps+ better insulation or radiative cooling materials that emit IR in the atmospheric window have the potential of being a disruptive technology for heating & cooling sector. Sky cool type panels are particular attractive, because radiative cooling works better at night and could take part of the after-office AC load off the grid. Solar anyway easily takes care of day time cooling needs.

BETTER (BATTERY) STORAGE OPTIONS REMOVES THE NEED FOR FRESH COAL FIRED POWER PLANTS AND ALLOWS TO INTEGRATE RENEWABLES

Electricity Demand

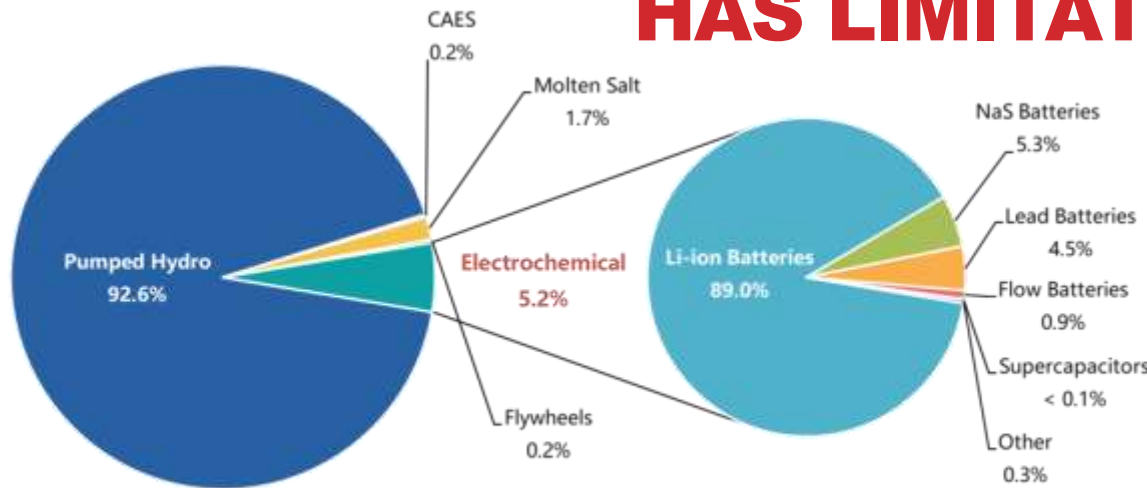


3 Weeks in Colorado (Xcel) 2003

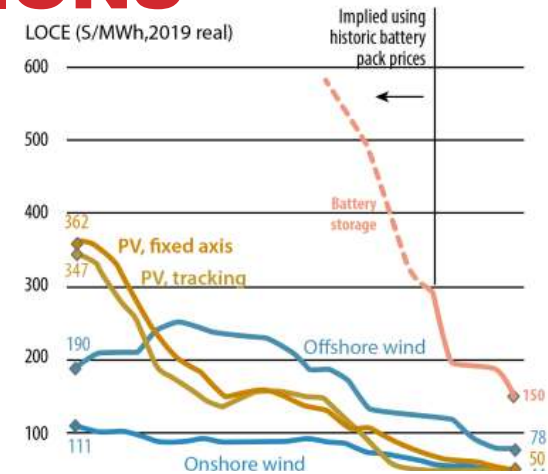
At 100% coal no interruptions coal has to run with ~50% capacity for most of the year in India because the demand peak in June is so high compared to the normal peaks.

This means recovering 100% cost by selling only 50% product. Technically, however, this is not possible because most of Indian coal is lignite & takes days to ramp up and down. Therefore, only ~50% of the installed capacity can be coal. The rest has to be something more responsive. Batteries are now the cheapest flexible load element.

GLOBAL ENERGY STORAGE HAS LONG BEEN STAGNANT BECAUSE PUMPED HYDRO HAS LIMITATIONS



Q1 2020

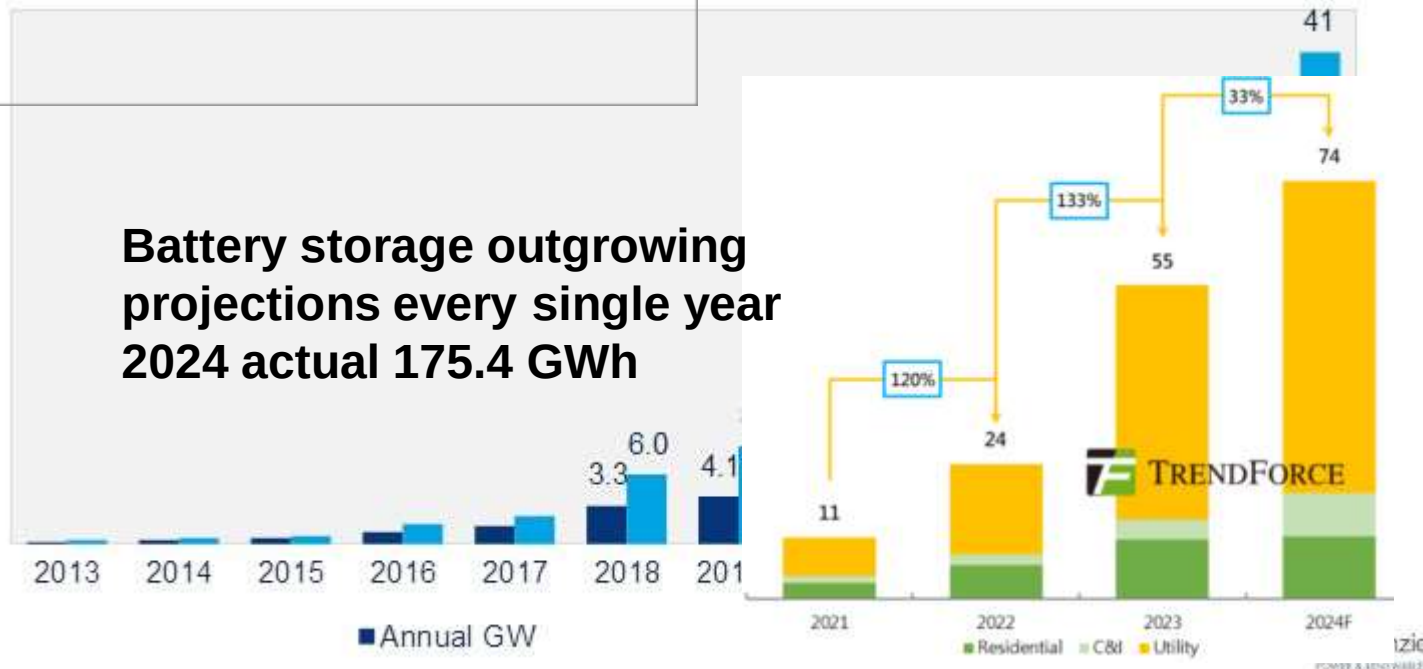


Gas peaker:
170\$/MWh

Battery: 150\$
MWh

Battery storage outgrowing
projections every single year
2024 actual 175.4 GWh

Global Annual Capacity GW / GWh



**WHAT
COVID
TAUGHT US**

CLEAN AIR IS POSSIBLE

For a brief while during lockdown the worlds most polluted cities accomplished their clean air goals.

Now we know that ...

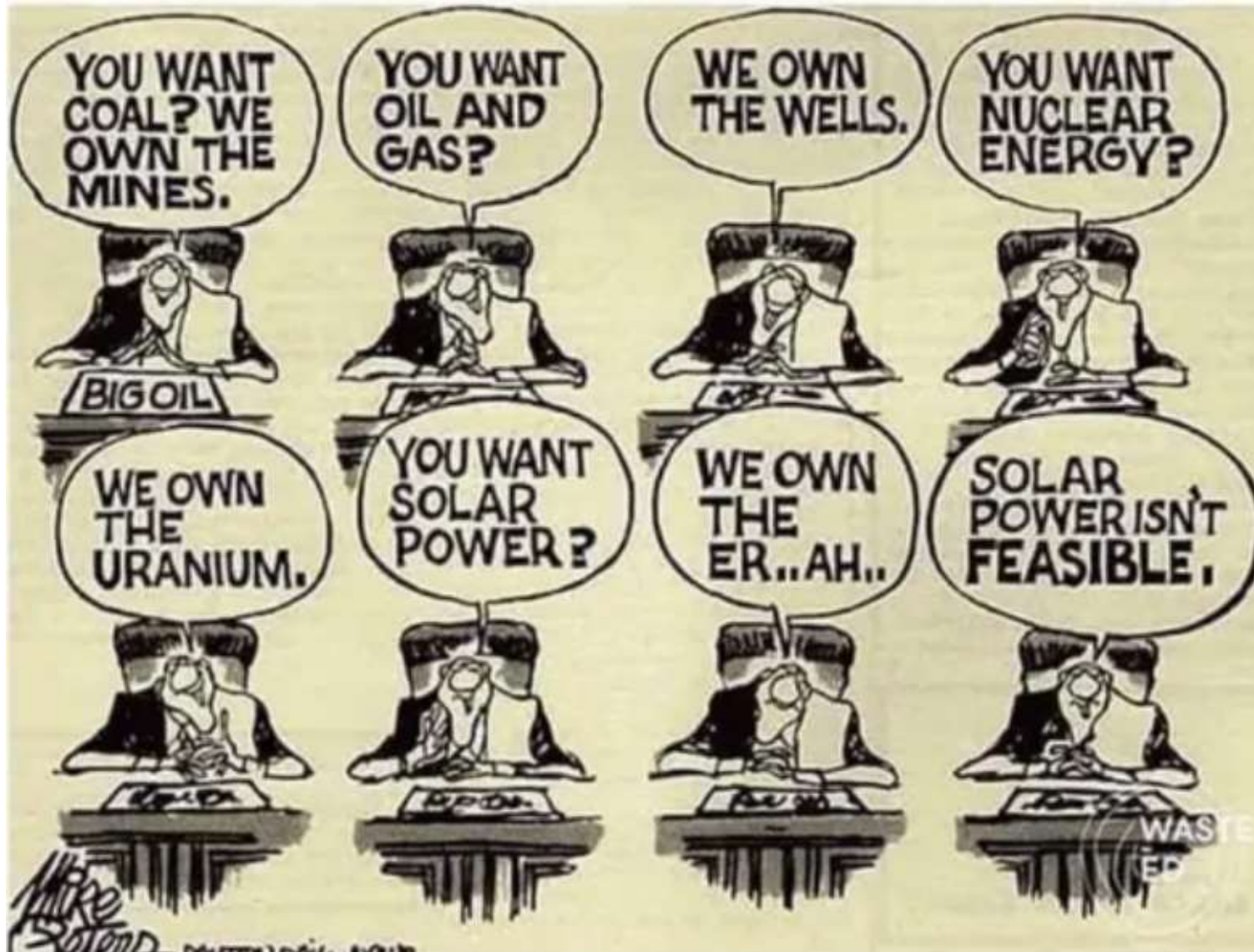
- 1) Making transport emission free through electric mobility
- 2) Making electricity emission free using renewable energy
- 3) Cleaning up the stack emissions from our industries



... can get the air quality in compliance with the NAAQS in most places in India. Our task is to figure out what it would take to make the necessary changes.

- What is stopping people who could cut their transport cost to 1/5th of their present cost or triple profit by switching to an electric 2 or 3 wheeler from making that choice?
- Why is coal still being favoured during tendering when the renewable electricity cost half?
- Why are people easily fooled into making choices that don't benefit them? (see e.g. discussion about nuclear in India, nuclear costs 20 INR per unit who wants to pay for that is the least cost option?)

CLIMATE CHANGE MITIGATION IS POSSIBLE



TAKE HOME MESSAGE

Battery storage has become big enough (200 MW = 1 Block of coal power) and cheap enough to:

- 1. Do away with night consumption incentives**
- 2. Accomplish grid stability with 40% lower installed peak capacity and with less flexible loads (Hydropower and natural gas)**
- 3. Integrate 100% renewables in the long run while giving end consumers lower cost**

The problem is renewable & storage are not monopolisable in the same way fossil fuels are. There is high competition so profit margins are small & consumers benefit.

Compressed biogas (46 INR/kg production cost) pays >80% of the total to your own farmers and laborers not to few rich people and a handful of engineers & scientists. Oil & gas extraction is very automated.

Estimated

Direct and Indirect Jobs

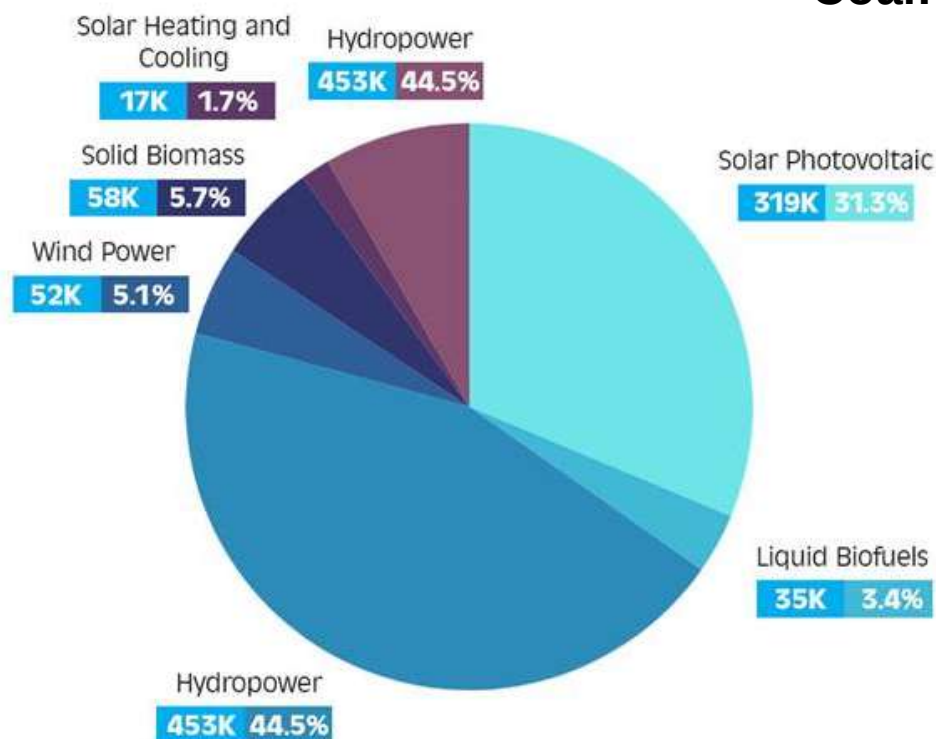
In Renewable Energy in India

(2023)



ONG: 110k

Coal: 337k +1200k informal



Source: IRENA's Renewable Energy and Jobs Annual Review 2024